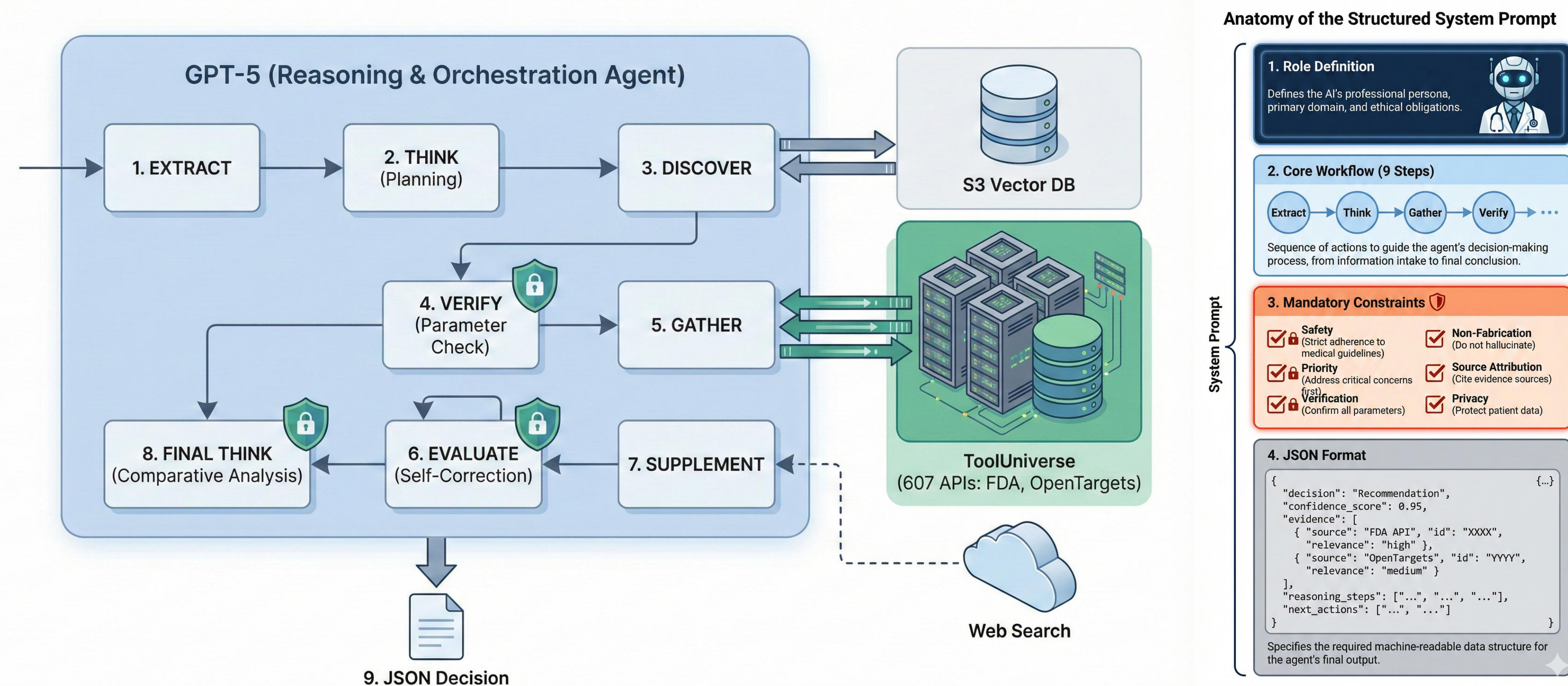


Motivation & Challenge

- ❑ **Context:** Precision therapeutics requires reasoning over complex patient data and 600+ clinical tools.
- ❑ **Gap:** Existing LLMs suffer from hallucinations and lack rigorous verification in high-stakes medical decisions.
- ❑ **Goal:** To build an agentic system that prioritizes clinical safety and logical consistency.

Methods



- ❑ **Safety-First Design:** The "Lock Icons" in the diagram represent Mandatory Verification Gates. The model cannot proceed without validating parameters (Step 4) or conducting a comparative analysis (Step 8).
- ❑ **ToolUniverse Integration:** Unlike generic agents, our system deeply integrates with ToolUniverse, pulling high-quality structured biomedical data (Step 5) rather than relying solely on parametric knowledge.
- ❑ **Adaptive Feedback Loop:** The logical flow allows for self-correction. If evidence is insufficient at Step 6, the system dynamically loops in Supplement (Step 7) to bridge information gaps.

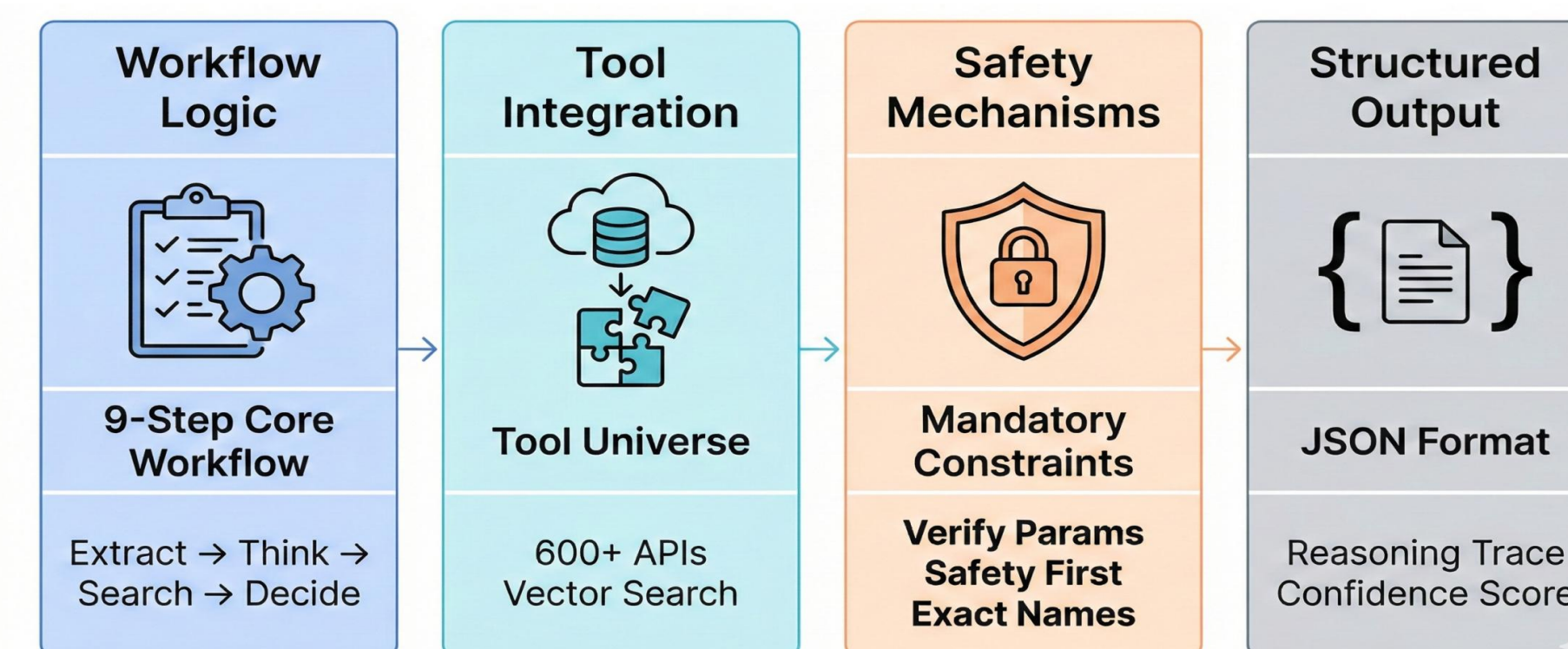
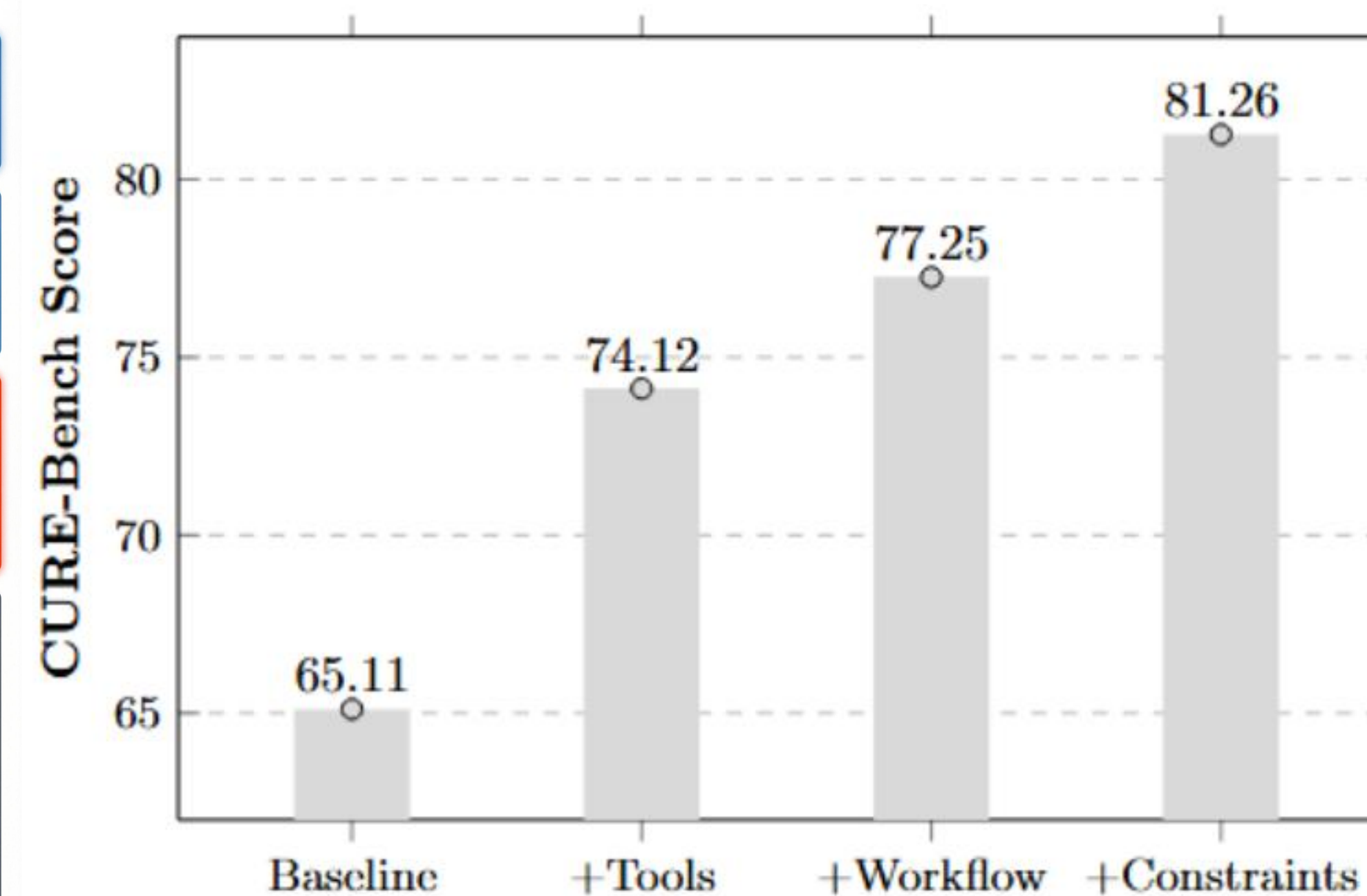
Conclusions

- ❑ **Enforced Safety** Mandatory verification checkpoints in our workflow eliminate hallucinations and ensure strict clinical compliance.
- ❑ **Precise Grounding** Prioritizing professional APIs (ToolUniverse) over generic web search guarantees accurate, evidence-based reasoning

Contributions

- ❑ **Structured Workflow:** A Multi-stage "Check-then-Act" pipeline with mandatory verification checkpoints.
- ❑ **Hierarchical Tool Use:** A strategy prioritizing professional databases (ToolUniverse) to ensure data quality.
- ❑ **Constraint-Guided Reasoning:** Explicit prompts forcing comparative analysis (Pros/Cons) for every therapeutic option.

Experiments and Results



- ❑ **Impact of Tool Retrieval (+9.01):** Transitioning from a vanilla LLM to a tool-augmented agent significantly reduced hallucinations, particularly in factual questions regarding drug ingredients and dosages.
- ❑ **Impact of Structured Workflow (+3.13):** Implementing the Multi-stage pipeline (Extract-Think-Decide) ensured that the model did not skip critical reasoning steps, improving performance on multi-hop reasoning tasks.
- ❑ **Impact of Mandatory Constraints (+4.01):** The final boost, which secured our high ranking, is attributed to the "Mandatory Safety Checks." By forcing the model to verify parameters (Step 4) and perform comparative analysis (Step 8), we eliminated safety-critical errors that standard agents missed.

More Information



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Main Reference: *Gao, Shanghua, et al. "TxAgent: An AI agent for therapeutic reasoning across a universe of tools." arXiv preprint arXiv:2503.10970 (2025).*